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An IoT-Enabled Growth Management System with Deep Learning-Based Monitoring for Volvariella volvacea Mushroom Cultivation in Control Environments

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An IoT-Enabled Growth Management System with Deep Learning-Based Monitoring for Volvariella volvacea Mushroom Cultivation in Controll Environments

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¹⁶

ABSTRACT

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¹⁹The nutritional profile and medicinal properties of the Volvariella mushrooms make them highly demanded in
²⁰Malaysia. However, with increasing demand, traditional methods frequently struggle to maintain the ideal
²¹environmental conditions necessary for optimal growth, leading to variable yields and quality of the products. To
²²overcome these challenges, we develop a system of an internet of things (IoT) based environmental control system
²³combined with a deep learning-based growth-stage monitoring system for Volvariella culture. The suggested IoT system
²⁴uses the ESP32 microcontroller to communicate with the DHT22 sensor for measuring temperature and humidity, the
²⁵MQ135 sensor for measuring carbon dioxide (CO₂) and the HW-038 sensor for detecting the water level. Actuators like
²⁶foggers, water pumps, heaters and ventilation fans are used by the system to continuously monitor and automatically
²⁷control critical environmental parameters. Optimum cultivation conditions were maintained in the range of 28–35 °C,
²⁸75–95% RH and CO₂ level less than 1000 ppm. A YOLOv8 deep learning model was trained to detect the mushroom
²⁹growth stage, using mushroom images with annotations of four stages: Tiny, button, egg and mature. The experimental
³⁰results showed that the proposed environmental control system was able to keep the target cultivation parameters within
³¹the target range for about 85% of the cultivation period. The overall classification accuracy of the growth-stage
³²recognition model was 93.12%, macro-average F1 score was 87.9%, and the mean average precision (mAP@0.5) was
³³0.78. Real-time monitoring and control were achieved using the Blynk mobile application, which allowed remote access
³⁴and saved about 90% manual labor. Passive ventilation effects were observed, but the system provided generally good
³⁵growing conditions and better cultivation reliability in the face of minor fluctuations in the environment. The use of
³⁶smart environmental control and vision-based analysis increases yield repeatability, quality, reduces human error and
³⁷manual operation. The presented results clearly show the capabilities of the proposed system to equip mushroom
³⁸cultivation with scalability and sustainability, thus improving the productivity of agricultural production and food
³⁹security.

4d. INTRODUCTION

⁴¹Volvariella mushrooms have high nutritional value in Malaysia, but their ideal growing conditions are difficult to sustain in
⁴²Malaysia, Volvariella mushrooms possess high nutritional value with the optimal growing conditions, which is difficult for the
⁴³farmers to maintain [1]. Traditional techniques have the drawbacks of uncontrollable temperature, humidity and carbon dioxide
⁴⁴(CO₂) and so it is difficult to get output that meets the desired criteria [2]. The farming systems are transitioning from traditional
⁴⁵manual operations to data-driven and autonomous systems thanks to the development of sensing technologies, automation, cloud
⁴⁶connections, and artificial intelligence (AI). Such developments are of particular significance for agricultural production sectors
⁴⁷where small variations in cultivation conditions can make a significant difference in production and product quality, and where
⁴⁸environmental management is of particular importance. Volvariella volvacea, known as the straw mushroom, is one of the most
⁴⁹commercially important and valuable edible mushrooms due to its nutritional properties. V. volvacea is a protein, vitamins,
⁵⁰mineral and antioxidant rich food source that can enhance food security and agricultural income in Malaysia and other countries
⁵¹in Southeast Asia [1, 2]. It has the potential for commercial production but is very dependent on proper environmental conditions
⁵²during the entire growing period. Mycelial activity, fruiting-body development and harvest quality are directly affected by
⁵³temperature, relative humidity and concentration of CO₂ [3]. Previous studies have shown that relatively high humidity (75–
⁵⁴95%) and moderate temperature (28–35°C) are generally favorable for the growth of the organism [3] [4]. Yet it is a significant
⁵⁵challenge to always maintain these environmental conditions, especially in tropical areas where the ambient conditions vary
⁵⁶often. This variation can stunt mushroom growth, make it easier to get contamination, and ultimately decrease mushroom

production. Production losses of up to 90% are reported in severe cases, and if production is mainly based on traditional monitoring and manual control of the environment [2,5].

Effective environmental management will thus be heavily dependent on the farmer's awareness of and capacity to respond promptly to environmental changes. Failures in taking corrective action put mushrooms in unsuitable growing conditions, which can lead to inconsistent growth performance and productivity [6]. The limitations highlight the need for more complex cultivation systems that are able to automatically respond to environmental changes and continuously monitor the environment. The IoT has ushered in a new era of possibilities for precision agriculture by connecting a network of sensors, embedded devices, and cloud-based communication tools, allowing for real-time monitoring of the environment. Applications of IoT technologies for mushroom cultivation have been studied, especially for the use of wireless sensor networks and low-cost microcontroller boards to monitor the temperature and humidity of mushroom growing [7] and [8]. These systems enhance visibility of the environment and help monitor it from a distance, but for many these are mostly data collection and display functions and not autonomous systems for managing the environment. In addition, the concentration of carbon dioxide, which is relevant to the respiration and fruiting-body formation of the mushrooms, is often not included in the environmental monitoring programs [9]. Consequently, the moment-by-moment adjustments of environmental conditions, such as changes in temperature and light intensity, are often left to the growers, which means that existing "smart" cultivation solutions are not practical. Meanwhile, AI and deep learning tools have gained prominence in the agricultural monitoring and decision-making processes. In object detection and classification and crop monitoring, computer vision (CV) techniques using convolutional neural networks (CNNs) have shown promising results. YOLOv8 (You Only Look Once version 8) has been a topic of significant interest due to its fusion of impressive detection accuracy, quick inference time, and computational efficiency [10]. Crop disease detection and estimation of yield, as well as agricultural quality assessment, have been successfully achieved using deep-learning methods. However, mushroom cultivation has not yet been widely adopted for their use, especially in the context of automatic growth stages recognition as part of environmental control systems. Previous studies have demonstrated the potential to use CNN for mushroom monitoring applications based on image analysis [8] and CNN/YOLO for mushroom detection [11] and [12]. However, these studies tend to be limited to visual recognition tasks without any consideration for environmental management, leading to cultivation systems with crop monitoring and environmental control as independent functions.

This separation is a major constraint as mushroom growth is known to be affected by the environment. A cultivation system that tracks mushroom development, but fails to consider the environmental status, gives only partial information for decision making. In turn, the environmental automation system, not to mention the intelligent cultivation management, cannot be carried out completely without crop-development awareness. Thus, there is a clear gap in the research of establishing an integrated framework that integrates environmental monitoring, automatic environmental regulation, cloud supervision and real-time mushroom growth-stage recognition on a single intelligent cultivation platform. From this perspective, this study proposed an intelligent growth management system for cultivating *Volvariella volvacea*, which is a combination of the environmental conditions monitoring system based on IoT, the automatic environmental management system, the remote supervision system based on the cloud computing system, and the growth stage recognition system based on YOLOv8. Temperature-humidity sensors DHT22, carbon dioxide sensors MQ135 and water-level sensors HW-038 are being used continuously to measure environmental conditions. The ESP32 microcontroller processes sensor data and autonomously adjusts cultivation conditions to favourable values by controlling foggers, ventilation fans, heaters and water pumps [13]. At the same time, the mushroom images are captured with a webcam, and processed using a mushroom image analysis system, which uses a deep-learning algorithm, YOLOv8, created by the Roboflow platform and trained using Google Colab [10]. The model has four mushroom growth stages: Tiny, button, egg and mature. The proposed framework is different from the typical method, where environmental monitoring and crop assessment are separated, and integrates both in a single smart cultivation platform. The system is built to provide real-time monitoring through the cloud-based Blynk system, which offers continuous, remote supervision of cultivation settings, as well as automatic environmental management and real-time growth stage identification. The proposed framework will combine the benefits of environmental automation and computer vision-based crop monitoring, with the goal of improving cultivation consistency, reducing labor requirements, increasing operational efficiency, and allowing for better informed cultivation and harvesting decisions. Lastly, this research enhances data-driven, intelligent, and sustainable mushroom cultivation systems.

4H. METHODOLOGY

48.2.1 Overall System Architecture.

The proposed intelligent growth management system integrates the IoT technology, automatic environmental control, cloud communication and computer vision based on the deep learning technology to develop the cultivation of *Volvariella volvacea* in a single system. The system provides a stable cultivation environment and manages mushroom development during the cultivation process. The proposed framework differs from the traditional cultivation practices, which are split into environmental management and crop assessment, because in the proposed cultivation approach both aspects will be performed using a single platform, allowing autonomous operation of the system and minimizing the manual supervision. The overall architecture consists of three interrelated subsystems: (1) environmental sensing and monitoring, (2) automated environmental control, and (3) intelligent growth-stage recognition. The ESP32 microcontroller is used as the main CPU that controls data collection, environmental management, data communication, and system control. The environmental sensing subsystem is permanently measuring the most important cultivation parameters such as temperature, relative humidity, CO₂ concentration, and the level of water. The DHT22 sensor is used for measuring temperature and humidity, while the MQ135 sensor is used to measure CO₂ concentration and air quality in the cultivation chamber. A HW-038 water-level sensor is used to measure water level in the fogger reservoir. The ESP32 processes the sensor data locally and uploads the processed data to the cloud platform for real-time visualization and data storage.

Automated subsystems of environmental regulation function with a closed loop feedback mechanism. The subsystem is based on ultrasonic foggers, ventilation fans, PTC heaters and water pumps that are linked to a relay control module. A series of

1 cultivation thresholds is predefined and taken as a reference for the various sensor readings, thereby enabling the system to
2 automatically react when the environmental conditions exceed the defined range of operation. This will allow for timely
3 corrective actions and help to keep the mushroom environment stable and conducive to mushroom growth and development.
4 Environmental data are sent to the Blynk Cloud platform using Wi-Fi for remote access and real-time monitoring. Users can
5 check on environmental readings via mobile and web-based dashboards and see the status of the actuators while also being able
6 to see the example in real-time. Farmers may evaluate and analyze cultivation trends and make better management decisions
7 based on the past environment information they have recorded by utilizing the cloud platform, which also makes it easier to
8 access. In addition to environmental monitoring and control, intelligent visual monitoring subsystems are also integrated to
9 automatically assess the growth stage of mushrooms. A YOLOv8 deep learning object detection and classification model is used
10 to process images from the webcam placed in the cultivation chamber for efficient real-time object detection and classification.
11 The model defines four pre-defined growth stages of *V. volvacea*, Tiny, button, egg, and mature. Detection results, such as
12 mushroom class labels, confidence scores and bounding boxes are created in real time and used to give visual feedback on
13 mushroom development and harvest readiness. The use of computer vision has revolutionized the way growers manage their
14 crops, allowing them to make better use of technology to track crop progress rather than relying on manual methods alone.
15 The proposed mushroom cultivation system is an intelligent system, as shown in Fig. 1. The system is divided into three
16 functional layers: sensing layer, control and communication layer, and intelligent analytics layer. The sensing layer collects
17 information from the cultivation chamber, and the control and communication layer processes the data measured by the sensors,
18 regulates the environment, and controls the connection with the cloud. The smart data analysis layer identifies growth stages
19 based on the YOLOv8 model and delivers visualization analysis of cultivation information. These layers are combined in one
20 system which will enable control of the environment and offer an automatic crop assessment. This integration enables the real-
21 time monitoring of cultivation conditions, autonomous environmental control, and data-driven decision-making, thus boosting
22 cultivation efficiency and fostering sustainable smart mushroom cultivation practices.

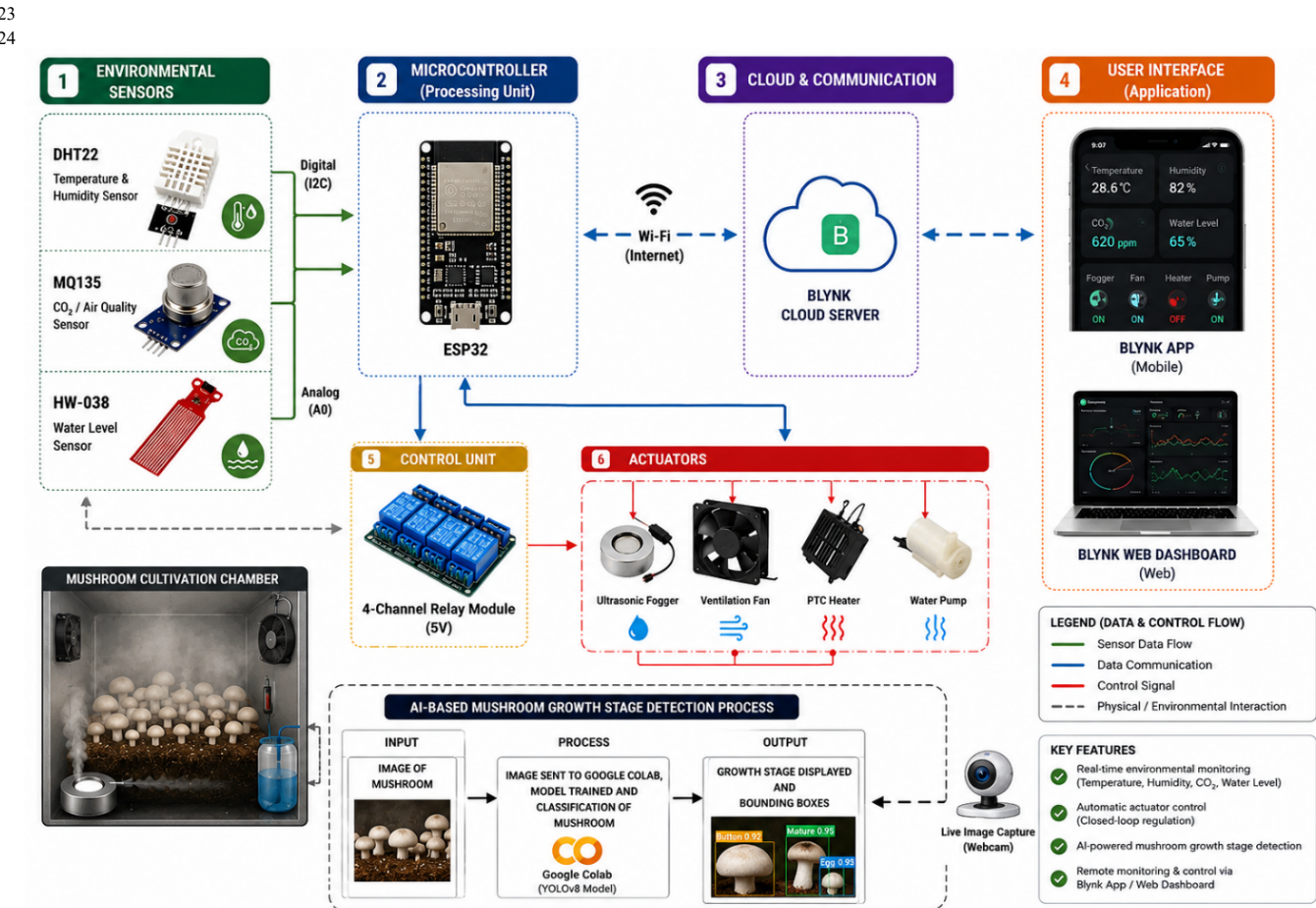


Figure 1. Overall intelligent mushroom cultivation system architecture.

2.2.2 Environmental Monitoring and Control System

The environmental monitoring and control subsystem was designed to keep the cultivation environment stable during the entire growth cycle because mushroom development is very sensitive to environmental stability, it is important to monitor and manage the environment on a continuous basis to achieve consistent development, high yields and product quality. Mycelial growth and fruiting body formation may be inhibited, depending on temperature, humidity, and concentration of CO₂, or water availability. In order to tackle these challenges, an environmental management framework based on IoT was designed, which will help to monitor and automatically regulate the environment in the cultivation chamber in real time. The low-power

consumption of the ESP32 microcontroller, along with its built-in Wi-Fi connectivity, was chosen for its use as the central processing unit and communication unit for this project, as it is well suited for real time IoT applications. The monitoring subsystem comprises of three sensor modules, which can measure critical environmental parameters continuously. The temperature and relative humidity are measured by a DHT22 sensor that is reliable within a broad operating range. The sensor can detect temperature $-40^{\circ}\text{C}\sim 80^{\circ}\text{C}$, humidity 0%-100%RH, which can be used to detect the temperature and humidity of the atmosphere in mushroom-growing rooms. An MQ135 gas sensor is used to measure carbon dioxide concentration and air quality inside the chamber, and the system can detect high CO_2 concentrations that can adversely affect the respiration and fruiting body development of mushrooms. Fogger reservoir is also equipped with an HW-038 water-level sensor to always have enough water for holding the humidification process.

The ESP32 microcontroller continuously reads and calculates sensor data and uses the results to determine whether the environmental control devices are operating. The control subsystem is composed of an ultrasonic fogger, a PTC heater, a fan for ventilation and a water pump, all linked by a 4-channel relay module. Closed loop control of the environment is achieved by comparing real-time sensor information with set cultivation thresholds. If the environmental condition is outside the desirable range, the corresponding actuator will be automatically turned on to make the environment suitable for cultivation. For instance, the ultrasonic fogger can humidify the air when the humidity measured is below the desired level, and the PTC heater is turned on when the air requires extra heating. Ventilation fan ensures superior air circulation and helps to prevent the buildup of excessive CO_2 , while water pump automatically refills the fogger reservoir when water reaches a low level.

The data from sensors and actuator states is stored on the Blynk Cloud platform on the ESP32 wireless communication interface to increase accessibility and flexibility of the operations. The mobile and web-based dashboards on the cloud platform enable users to access cultivation parameters remotely and in real time and to track system performance. The platform also includes historical data storage and retrieval capabilities, enabling growers to analyze cultivation trends and make data-driven decisions based on the data stored. The platform also has historical data storage and retrieval options, allowing growers to review cultivation trends and make informed management decisions based on the data recorded. The architecture of the proposed environmental monitoring and control subsystems is shown in Fig. 2. All the environmental data parameters collected inside the cultivation chamber are sent to the ESP32 microcontroller for further processing and decision making. The controller uses the measured data to calculate the control signal for the actuators, which controls the humidity, temperature, air circulation and water supply. Meanwhile the environmental data is being sent to the Blynk Cloud platform, where it can be monitored remotely and recorded. The proposed architecture integrates the sensing, communications and automated control functions to create a closed loop environmental management system that can react continuously to environmental changes and keep the cultivation environment stable throughout the cultivation period.

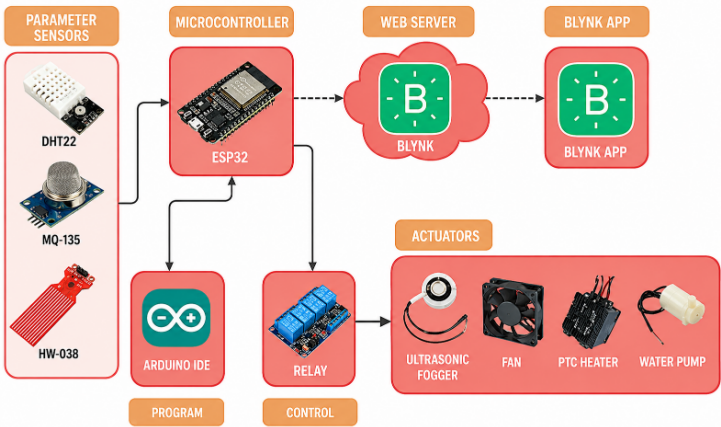


Figure 2. Environmental monitoring and control block diagram integrated with ESP32 and Blynk Cloud platform

Closed loop control of environmental parameters was applied to keep the cultivation conditions within the desired range during the whole growing period of *Volvariella volvacea*. The proposed cultivation method constantly monitors the environment and automatically takes corrective measures if the environmental conditions fall outside the desired operating range, as opposed to traditional cultivation methods, which rely heavily on manual observation and action. This system combines all the advantages of real-time sensing and autonomous actuator control, thereby ensuring the stability of the cultivation environment while minimizing the need for constant human monitoring. The control process starts by receiving the temperature, relative humidity, water-level and CO_2 measurements made by the environmental sensing subsystem. The measurements are then taken by the ESP32 microcontroller and compared to a set of thresholds defined based on the growth conditions for *V. volvacea*. The evaluation results are used by the controller to decide whether certain adjustments of the environment are needed and to switch the appropriate actuators on when needed, to restore favourable cultivation conditions. The ventilation subsystem is activated when inadequate air circulation is detected, or too much gas is detected within the cultivation chamber to maintain adequate air quality within the cultivation chamber. The continuous assessment of the environment enables the system to be dynamic and adjust cultivation conditions optimally, resulting in a stable microclimate and reduced energy use and actuator operation.

1 The operation flow of the proposed environmental regulation algorithm is shown in Fig. 3. Environmental measurements
2 are taken continuously, following system start-up, and forwarded to processing unit for analysis. All environmental parameters
3 are checked against threshold limits and appropriate control measures are implemented when they exceed the thresholds.
4 Environmental conditions are evaluated after every corrective action to see if additional regulation is needed. This is a recursive
5 process and repeated until the parameters are within desired operating range. Once the cultivation is stabilized, the actuators
6 are switched off, and the most up-to-date environmental information is transmitted to the cloud platform for visualization,
7 storage and remote monitoring. There are several practical benefits to the closed-loop control strategy. It helps in providing
8 timely response to environmental disturbances without manual interventions. Second, it reduces fluctuations in important
9 environmental parameters which could adversely impact mushroom growth and development. Third, it reduces the needless
10 activity of actuators, which in turn helps to extend the lifespan of the equipment and save energy.

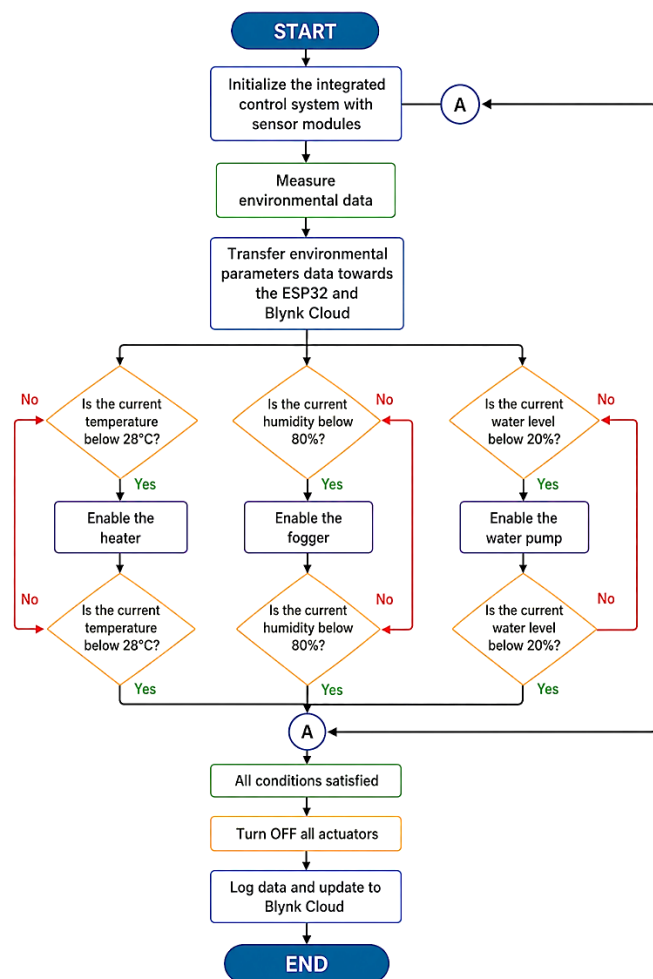


Figure 3. Flowchart of the closed-loop environmental monitoring and actuator control algorithm implemented in the ESP32 microcontroller

15 The environmental regulation algorithm works based on the set threshold values. The first evaluation is temperature because
16 temperature is a major factor in mushroom development. If the temperature measured is below 28°C, the ESP32 turns on the
17 PTC heater via the relay control module to heat the temperature of the chamber. The temperature is automatically measured
18 until a desired level is reached. The relative humidity is controlled in a similar way. The ultrasonic fogger automatically
19 activates as the measured humidity is reduced below 80%, to bring the humidity level back up in the cultivation chamber.
20 Humidity readings are then re-evaluated to make sure the proper environment is restored. The HW-038 water-level sensor is
21 used to simultaneously monitor the water level in the fogger reservoir. If the water level is below 20%, the water pump will
22 automatically start as soon as the measured water level is low, to ensure the reservoir water is topped up and the humidification
23 system continues to operate. By continuously checking water-level status, lack of water conditions that can affect the
24 environmental control system are avoided. If all the environmental parameters meet the set operating conditions, the control
25 system turns off the heater, fogger and water pump, preventing waste of energy. The new measurements from the environment
26 and the status of the actuators are then recorded and sent to the Blynk Cloud platform, which can be visualized real-time,
27 supervised remotely and analyzed in the past. The proposed system will achieve the purpose of maintaining environmental
28 stability, cultivating regularity and providing autonomous smart farming operations by adopting a closed system operating
29 mechanism.

1 The circuit implementation of the proposed environmental control system is given in Fig. 4. The schematic shows how the
2 sensing modules are connected to the relay drivers, voltage regulation, and environmental actuators. The ESP32 controller
3 processes the readings received from the DHT22, MQ135 and HW-038 sensors and produces relay control signals to control
4 the operation of the actuators. Stable voltage regulation is provided to low voltage components using LM2596 voltage regulator
5 modules, which include the power regulation. The power regulation is included with LM2596 voltage regulator modules used
6 to provide stable power to low voltage components, improving overall system reliability. In addition, the modular circuit design
7 allows for easy troubleshooting, maintenance and flexibility for future upgrades and expansion.

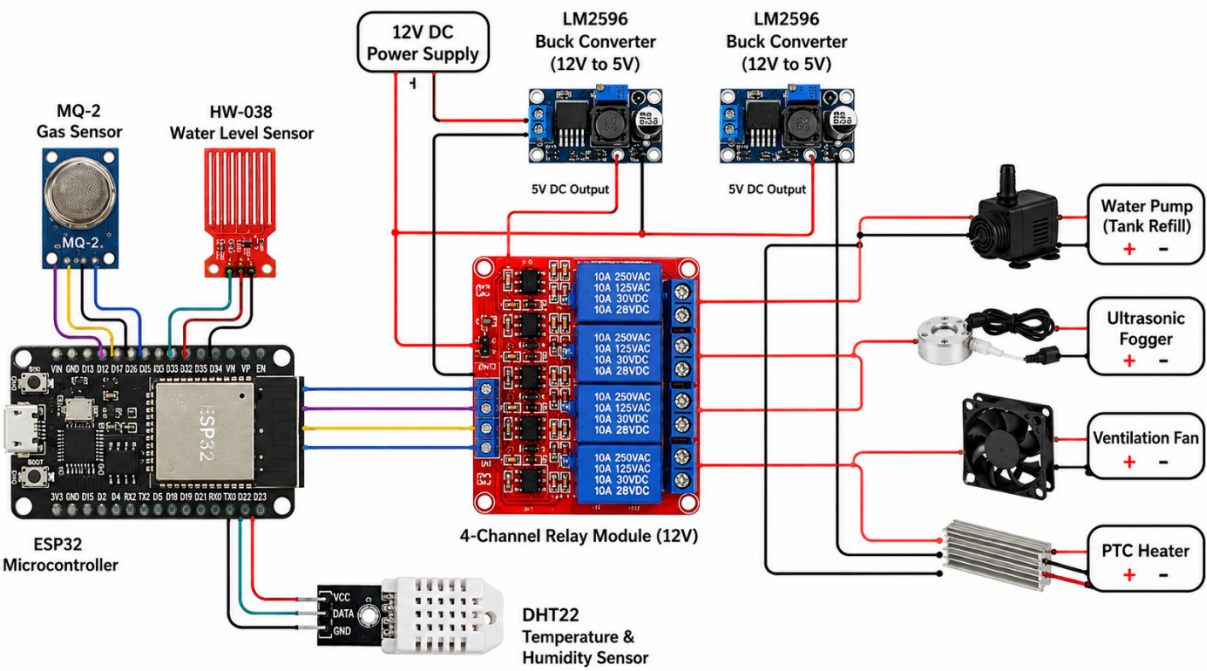


Figure 4. Circuit schematic of the ESP32-based environmental monitoring and automated environmental control system.

11 2.3 Hardware Assembly and Experimental Setup

12 The mushroom cultivation system under proposal was designed and implemented in the specially created mushroom cultivation
13 chamber, which is designed to be used for continuous environmental monitoring, automatic regulation of the environment, and
14 observation of the growth stage. This prototype was built with the architecture set out in the previous section where the sensing,
15 control, communication and computer vision modules were combined into an operational platform. This integrated solution
16 enables environmental management and monitoring of the crops simultaneously, providing a viable mushroom farming
17 framework with intelligence. The microcontroller that is at the heart of the system is the ESP32. This is a microcontroller that
18 can be used for both processing and communicating. The controller controls and manages wireless communication with the
19 cloud platform, collects the sensor data, and executes environmental control decisions. Temperature, humidity, gas
20 concentration and water level are measured by the DHT22 temperature and humidity sensor, MQ135 gas sensor and HW-038
21 water level sensor, respectively. The sensing modules were installed in a strategic position for obtaining maximum
22 representative monitoring of cultivation internal environment and to get more reliable monitoring of internal environmental. A
23 relay control module is used to provide the electrical connection between the ESP32 and the environmental actuators, thus
24 controlling the environment. Based on the decisions made by the closed-loop control algorithm, the relay module drives the
25 ultrasonic fogger, PTC heater, ventilation fan and water pump. With this setup, changes to the environment can be made
26 automatically without user action. The electronic subsystem is realised with the use of voltage regulation with the help of adding
27 the buck converter modules (LM2596) to the electronic subsystem, which make up its stability and reliability. These will
28 provide a regulated power supply for the microcontroller, sensing devices and communication circuits, helping to minimise
29 voltage fluctuations that could affect the accuracy of the measurements or operation of the system. Good power regulation is
30 particularly important in long-term cultivation systems, where continuous operation is a must.

31 The inside view of the proposed cultivation prototype (hardware integration) is depicted in Fig. 5(a). The figure presents
32 the configuration of the sensing modules, the control electronics, the communication elements and environmental actuators of
33 the cultivation chamber. Components were sited, electrical routing was carefully done and subsystems isolated to assure
34 electrical safety, maintainability and reliability of operation. These factors are crucial for long-term stable deployment in real
35 cultivation. The entire experimental prototype is presented in Fig. 5(b). The environmental parameters inside the cultivation
36 chamber were set during the experimental evaluation within the range of nutritional parameters recommended for the growth
37 of the mushroom species *Volvariella volvacea*, namely temperature between 28°C and 35°C, and relative humidity between

175%–95%. Environmental regulation was carried out continuously throughout the growing period to prevent fluctuations and to assure uniform growing conditions. This successful deployment of prototype proves the feasibility of the integration of IoT based environmental automation, cloud-based monitoring and intelligent growth-stage analysis in one cultivation platform. The results obtained demonstrate that the proposed method can be a viable solution for smart mushroom cultivation systems and decrease the need for manual monitoring and control of the environment.

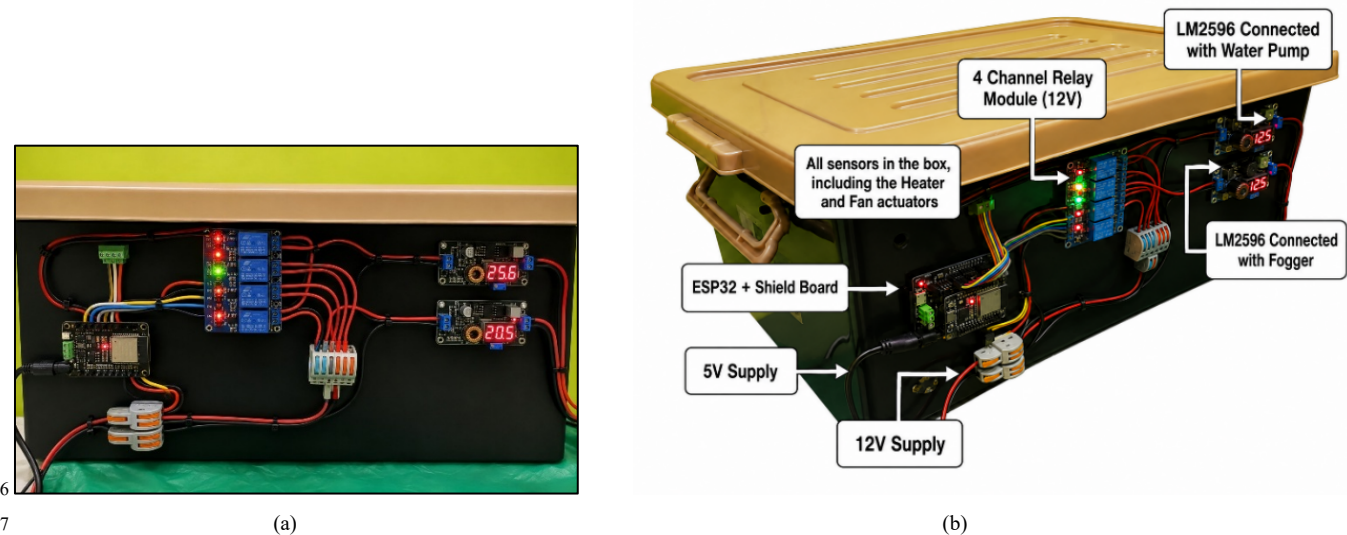


Figure 5. (a) Internal hardware integration of the proposed intelligent mushroom cultivation prototype. (b) Complete hardware prototype and experimental implementation of the intelligent mushroom cultivation system.

12.2.4 Dataset Acquisition and Annotation

The mushroom growth-stage detection model was trained and evaluated using a dedicated image dataset. The mushrooms were captured within the cultivation chamber using a webcam that was set up to record the mushroom growth process throughout the cultivation period. Data were gathered under actual cultivation conditions to ensure that the data set was representative of realistic variations in environmental conditions, lighting, mushroom orientation, and mushroom characteristics. This was done to make application to real cultivation environments more feasible. *Volvareia volvacea* was divided into 4 growth-stage categories by means of the observed morphological features. The categories included tiny, button, egg and mature which correspond to the main phases of a mushroom's life cycle observed during mushroom growing and harvesting. These categories were established to enable the model to detect them at the stage and to provide some useful information for monitoring cultivation and for harvest planning. After collecting images, the data set was uploaded to the Roboflow image annotation database for annotation and management. Each mushroom in the image was labeled with a rectangular bounding box and was assigned a class label representing the growth stage of the mushroom. The resulting annotation process produced a structured dataset which can be used for supervised learning object detection and classification. Detailed annotations were applied to make sure the uniformity throughout all growth stages and to enrich the training data.

To increase the complexity and diversity of the data set, and to build a robust model, a few data pre-processing and data augmentation methods were employed. Enhancements such as image size change, image normalization, horizontal image flipping, rotation, zoom augmentation and contrast adjustment were chosen for the following transformations. By providing the model with varying visual situations of the data, data augmentation allows the model to learn features that are less dependent on the illumination, scale, camera orientation and background appearance of the objects. Thus, the trained model can adapt well to the actual cultivation environment. The annotated data was then exported to be used with YOLOv8 and divided into 73% training, 21% validation and 6% testing, respectively. This partition method enabled the development of a systematic model, hyperparameter optimization and independent performance evaluation. The training subset was employed to learn the features and classification patterns of the training objects, and the validation subset was employed to prevent overfitting the model by tuning the model. The testing subset was only applied at the end after the model was finalized to give a true picture of the effectiveness of the model. The whole data preparation workflow is presented in Fig. 6, where the images were acquired and the dataset is then partitioned. The workflow shows the steps of the annotation/preprocessing, augmentation, and organization of the dataset for the development of the YOLOv8 model. These procedures are all linked together and add to the model generalization and mushroom growth-stage recognition reliability in real-world cultivation conditions.

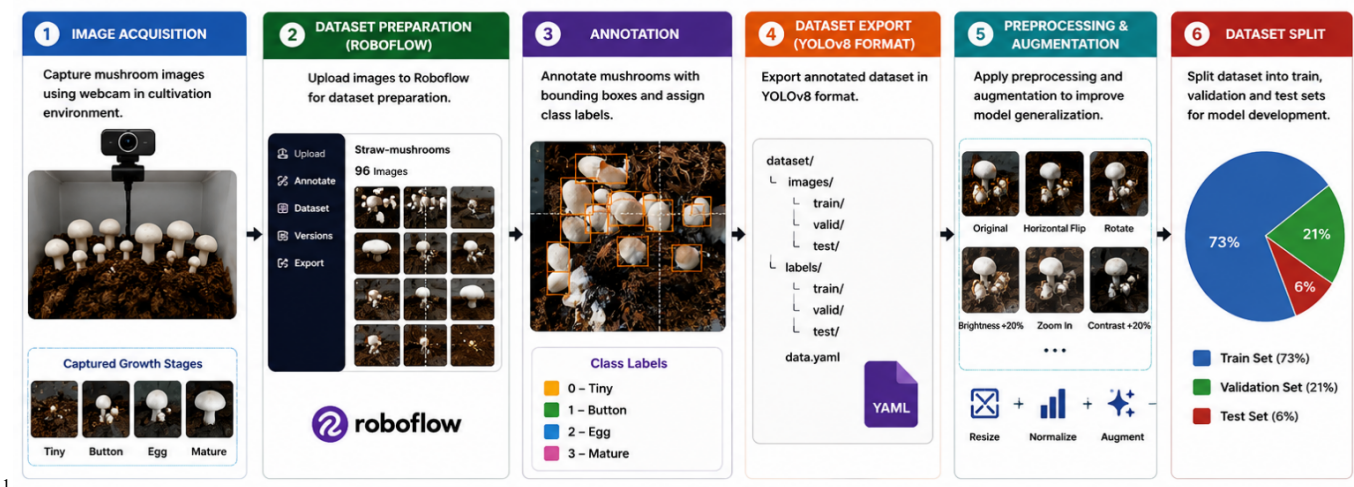


Figure 6. Workflow for mushroom image acquisition, annotation, preprocessing, augmentation, and dataset partitioning using the Roboflow platform

2.5 YOLOv8 Model Development and Training

The mushroom growth-stage recognition model is adopted because it has high accuracy in mushroom growth-stage recognition, and is high-speed inference, with low computational complexity. The capabilities of YOLOv8 make it suitable for use in real-time agricultural monitoring applications, where timely detection and efficient resource utilization can significantly impact the overall success of the operation. The training and testing of the model were conducted with GPU acceleration in the Google Colab environment, which cut down the computational time, and enabled efficient training and testing. The lightweight and resource-friendly nature of the pretrained YOLOv8n architecture made it an ideal choice for deployment on resource-constrained systems. Transfer learning was used to fine-tune the model weights obtained by pretraining on the mushroom growth-stage recognition task. This allows the model to leverage the feature representation that was learned from a large image set but also allows for a more tractable amount of training time and a more tractable convergence behavior.

Four mushroom growth-stage categories tiny, button, egg, and mature were prepared and loaded into the training environment of YOLOv8. The optimization of the model was carried out over 50 epochs of iterative forward and backward propagation. The objective of training was to minimize prediction errors and increase the accuracy of prediction; whereby model parameters were updated during training. The validation process was carried out throughout the training, to evaluate the convergence behavior, stability of the model and possible overfitting. Multiple quantitative performance measures were adopted to assess model performance, such as precision, recall, F1 score, mean average precision at intersection-over-union 0.5 (mAP@0.5). These metrics are complementary to each other in terms of information about the accuracy of detection, the reliability of the classification and about the effectiveness of the model, in general. Moreover, the confusion matrix was analyzed to explore the misclassifications between the four growth-stage groups and to locate the classes to be more difficult to distinguish. Hyperparameter tuning and several folds of validation were carried out for model accuracy, when required.

The entire deep learning process used in this study is depicted in Fig. 7. It begins with importing the data from the Roboflow platform, trains the model, validates the model, checks the performance of the model, and prepares the model for deployment. The systematic development process enables the reusability of the model, and the model performance evaluation is carried out throughout the training process. After successful validation, the trained model, YOLOv8 was deployed for monitoring mushroom growth stages in real-time. A webcam was installed inside the cultivation chamber, and pictures were taken every moment to be processed by the trained model for identification and classification of mushroom growth phases. The results of the detection, such as bounding boxes, confidence scores and predicted class labels were displayed using visualization functions from OpenCV for immediate feedback on mushroom development. The deployment mode allows for continuous monitoring of plant development and helps to timely implement harvesting decisions and further promotes the practical application of the proposed intelligent cultivation system.

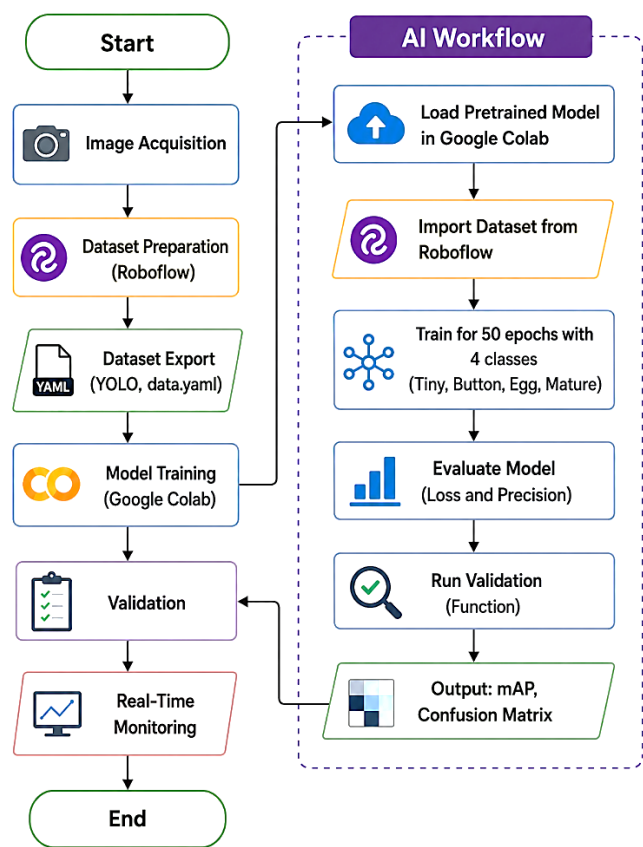


Figure 7. YOLOv8 model development, training, validation, and real-time monitoring workflow.

4.2.6 Experimental Evaluation and Performance Analysis

Detailed experiments were carried out to assess the effectiveness of both the environmental automation subsystem and mushroom growth-stage recognition subsystem based on YOLOv8. The evaluation's goal was to confirm the feasibility of the proposed intelligent cultivation framework to ensure stable cultivation environment and dependable monitoring of the growth stages in the cultivation process. The performance of the integrated IoT and deep learning system could be systematically validated by analysing the performance of the environmental regulations and the detection accuracy of the regulations. The design of the environmental automation subsystem was evaluated based on its ability to manage the cultivation parameters within the desired range. Throughout the growing period, important environmental factors, including temperature, relative humidity, CO₂ concentration, and water level were monitored. The operational characteristics of the actuators (heater, fogger, ventilation fan and water pump) were also measured in order to assess the responsiveness of the closed loop control strategy. A percentage for environmental compliance was calculated for each of the monitored parameters, defined as the percentage of time the monitored parameter remained within the desired range. This is a direct measure of the system's ability to keep the growing conditions favorable and reduce the fluctuation in the environment.

To measure the performance of the growth-stage recognition subsystem, common metrics for object-detection, namely the mean average precision at intersection-over-union (IOU) threshold of 0.5 (mAP@0.5), precision, recall, and F1 score were used. Precision, which is the ratio of correctly detected mushrooms to the number of mushrooms detected by the model; recall, which is the ratio of mushrooms relevant to an image dataset to the number of mushrooms detected by the model. The overall performance of the model is evaluated using the F1 score and the overall detection and localization performance for all growth stage categories is evaluated using the mAP@0.5. These metrics are used in conjunction with each other to provide a good indication of the model's classification and detection abilities. A confusion matrix was employed for the four pre-defined growth stages tiny, button, egg and mature, when analyzing the behavior of the models. Based on this analysis, authors were able to pinpoint misclassification patterns and offer more information about the model's weaknesses and strengths. Interclass confusion is important to remember because sometimes growth stages that are close together have similar visual characteristics and discrimination is more difficult. These results were then used to verify the effectiveness of the proposed intelligent cultivation framework through an analysis of the environmental monitoring results and the detection of the performance metrics of the objects. The assessment of the environmental regulation capability together with the growth-stage recognition accuracy make a comprehensive assessment of the overall performance of the system. The experimental results also confirm the possibility of using the environmental automation system based on the IoT technology together with the visual analytics based on deep learning for intelligent and data-driven mushroom farming application.

III. RESULTS & DISCUSSION

This section is an experiment of the proposed mushroom cultivation system with intelligent system under controlled cultivation environment. The assessment is carried out in four main aspects of the system performance: environmental regulation capability; mushroom growth progression; mushroom monitoring in the cloud; and usability of mushroom growth-stage recognition model based on YOLOv8. The relationship between the stability of the environment and mushroom development is discussed in detail as both are direct components of the overall effectiveness of the proposed mushroom cultivation system. It is designed to provide a reliable, data-driven system for mushroom growing with the added benefit of automatic environmental control and intelligent visual monitoring. The experimental outcomes provide valuable insights into feasibility, reliability and promise of applying IoT technologies with deep learning methods in smart cultivation systems.

3.1 Environmental Monitoring and Control Performance

The environmental monitoring and control subsystem was assessed for 9 days of cultivation, to determine how well the microclimate conditions required for growth of *Volvariella volvacea* could be maintained. The temperatures, relative humidity, carbon dioxide concentration and reservoir water level were recorded continuously throughout the experiment under both day and night operation. The environmental factors chosen due to their significant effects on mushroom development, fruiting body formation and overall mushroom productivity in cultivation. In Table 1 the environmental measurements recorded are summarized. Data collected were analysed to assess the responsiveness and effectiveness of the proposed closed-loop control strategy to maintain cultivation conditions within the recommended operation range. Actuator’s behaviour has also been analyzed to see if the control system could react in the correct manner in the face of changes in the environment. If the environmental regulation is consistent all over the mushroom culture period, it means the proposed automation system can minimize the reliance on manual operation and promote the positive environment for mushrooms growth and development.

Table 1. Recorded environmental parameters from day 1 to day 9 during mushroom cultivation

Day:Time	Temperature (°C)	Humidity (%)	CO ₂ (ppm)	Water Level (%)
D1 : Day	37.4	85	87	100
D1 : Night	35.8	85	87	100
D2 : Day	34.0	94	50	88
D2 : Night	33.4	83	48	90
D3 : Day	34.8	82	53	74
D3 : Night	32.5	85	54	100
D4 : Day	35.0	85	57	82
D4 : Night	34.9	83	63	89
D5 : Day	35.2	85	77	54
D5 : Night	35.0	83	42	94
D6 : Day	35.0	85	175	73
D6 : Night	35.0	81	64	60
D7 : Day	35.0	85	112	100
D7 : Night	36.0	85	143	81
D8 : Day	35.1	88	165	100
D8 : Night	35.2	85	186	85
D9 : Day	35.5	85	176	100
D9 : Night	35.3	84	129	100

The temperature attained during cultivation was kept within the recommended range of 28-35°C during most of the experimental period as shown in Fig. 8(a) with the help of the proposed environmental control system. The initial rise in temperature at the start of cultivation was 37.4°C, this is likely due to the accumulation of heat inside the chamber prior to the control algorithm reaching stable operating levels. The temperature was then slowly stabilized and kept within the desired range following this transient period, indicating the ability of the closed-loop control mechanism to effectively control the thermal condition. During the nighttime sampling period of day 3, a marked decrease in temperature was noticeable; the temperature of the chamber dropped to around 32.5°C. The control system, being programmed to a PTC heater through the relay-based control module, sensed this deviation and turned on the PTC heater to raise the temperature back towards the target operating range. The relatively short recovery period indicates that the environmental regulation algorithm is a very responsive one and has a compensation effect for the external environmental disturbance. Thermal environment is of vital importance in the cultivation of *Volvariella volvacea* since temperature plays an important role in the activity of mycelium, primordia formation and in the production of fruiting-bodies. Rapid changes in temperature could cause disruption in growth and have a negative impact on

1 mushroom production and quality. The temperature profile recorded over the 9-day culture period indicates that the system was
2 able to provide optimal and stable thermal environment during the culture period, which is conducive for mushroom cultivation.
3 The results showed that the proposed automated temperature regulation strategy was effective in controlling the temperature of
4 the mushroom cultivation system, thereby confirming its feasibility for mushroom farming applications.
5 The moisture profile shown in Fig. 8(b) shows constant moisture control during the entire cultivation process. Relative
6 humidity was maintained around the desired set point of 85% with only minor variations noted over selected nighttime periods.
7 Humidity levels were the lowest on day 3 with roughly 82% and on day 6 with roughly 81%. These are short-term reductions,
8 which may be a result of passive air exchange and natural fluctuations of the cultivation environment within the chamber. The
9 ultrasonic fogger subsystem functioned well to maintain the desired operating range humidity, even though there were these
10 fluctuations. The maximum humidity recorded was 94% on day 2 and all were within the acceptable range needed for healthy
11 mushroom development. This is significant as adequate moisture is needed to prevent dehydration and to promote the growth of
12 fruiting bodies. The control system performance to hold the environment within a certain range of humidity verifies the
13 appropriateness of the proposed environmental regulation framework to the application of mushroom cultivation.

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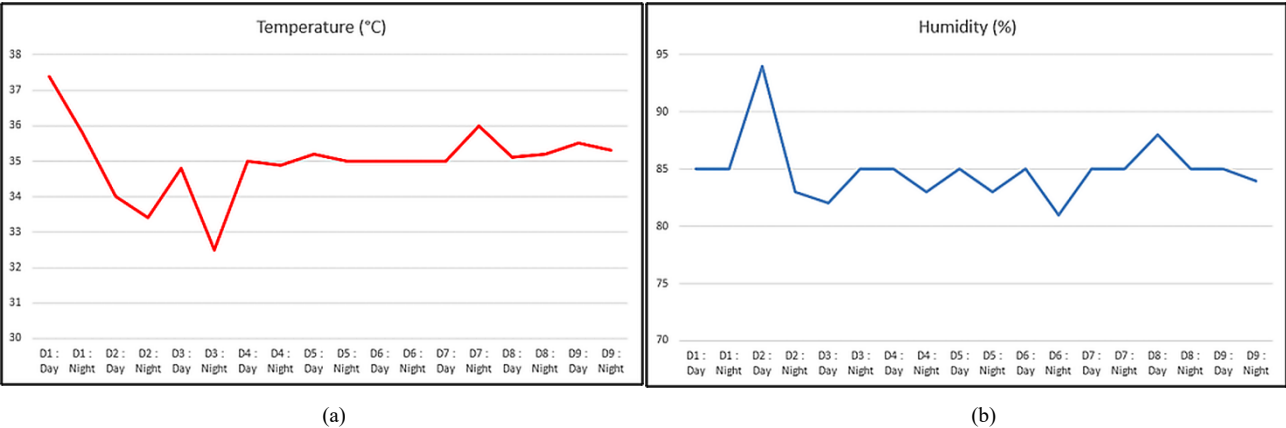


Figure 8 (a). Temperature regulation performance throughout the cultivation period. (b) Humidity regulation performance throughout the cultivation period.

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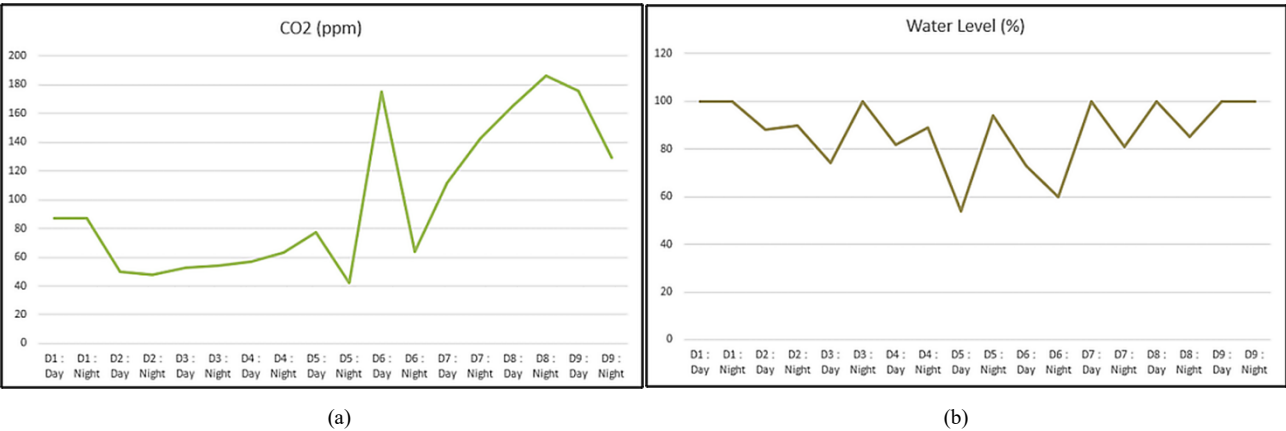


Figure 9 (a). CO₂ concentration profile throughout the cultivation period. (b) Water-level monitoring and automatic replenishment performance

23 In addition, this carbon dioxide concentration profile as shown in Fig. 9(a) further illustrates the effectiveness of the proposed
24 environmental management subsystem. The CO₂ levels stayed relatively constant and low (below 90 ppm) during the first six
25 days of cultivation, indicating good circulation of air inside the cultivation chamber. Concentration was gradually increased after
26 day 6, after adding yeast fermentation, which was deliberately introduced to produce CO₂ to mimic the biological gas production
27 during cultivation. Nevertheless, the maximum recorded value was 186 ppm, which is still below the usual value of 1000 ppm
28 linked to bad conditions for cultivation. No significant changes in the cultivation environment suggest that it stayed within a safe
29 operating range during the experiment. The reduction of CO₂ concentration was observed periodically along with the operation
30 of the ventilation subsystem, indicating the effectiveness of the gas accumulation management mechanism. The response
31 observed substantiates the ability of the closed loop control strategy to keep the air environment balanced even when more carbon
32 dioxide was added to the chamber. This regulation is especially significant because excessive CO₂ levels can delay normal
33 mushroom development, cause a decrease in growth and negatively impact fruiting-body formation. The results accordingly
34 indicate that the proposed system was able to maintain proper air quality and to incorporate the controlled biological processes
35 in the cultivation environment.

1 The performance of the AWMS through the cultivation cycle is shown in Fig. 9(b). The system can be used for continuous
2 monitoring with the HW-038 water-level sensor, which allows us to identify the depletion of the reservoir caused by an extended
3 fogger running time. The water level showed a significant drop on day 5, related to a higher humidification level required during
4 environmental regulation. As the water level was approaching the set-up, the control algorithm automatically switched on the
5 water pump to restore the water in the reservoir. After replenishment, the water level was stabilized and the water level during
6 the cultivation period was kept within the operating range. The automated replenishment mechanism operated successfully and
7 allowed for continuous functioning of the fogger and a direct contribution to stability of the humidity regulation process.
8 Humidification should be continuous and maintaining adequate levels of water in the reservoir is important, as a low level may
9 reduce the effectiveness of humidification and adversely impact the growth of mushrooms. The proposed system was designed
10 to automatically adjust water levels as needed, reducing the need for manual operation and enhancing the system's reliability.
11 The results show that the adopted water-level sensor and water level auto-replenishment system is effective and helps to make
12 the system more practical for long-term smart mushroom production.

13 3.2 Cloud-Based Monitoring and Remote Accessibility

14 One of the main goals of the proposed cultivation plan was to be able to provide information about the environment without
15 having to interact directly with the cultivation chamber. This was made possible by integrating with the Blynk.Cloud platform
16 that allowed real-time data transmission, visualization and storage of data from the environment collected by the ESP32
17 controller. The system integrates wireless communication and cloud services, enabling users to check the cultivation
18 environment remotely while keeping the environment supervised continuously. The web-based dashboard and mobile
19 monitoring interface designed for the proposed cultivation system are shown in Figures 10 and 11, respectively. Figure 10
20 shows a web-based dashboard for the proposed cultivation system, and Figure 11 shows a mobile monitoring interface for the
21 proposed cultivation system. Both platforms achieved displaying the environmental parameters in real time, such as the
22 temperature, the relative humidity, the percentage of water level and the air quality indicators. The system successfully sent
23 representative data of temperature 35.4°C, relative humidity 83%, water level 65%, and air quality index (AQI) value to the
24 cloud platform in real time during the operation of the system. These parameters have been successfully displayed, reflecting
25 the effectiveness of the communication system and its capacity to deliver regular environmental feedback to users.

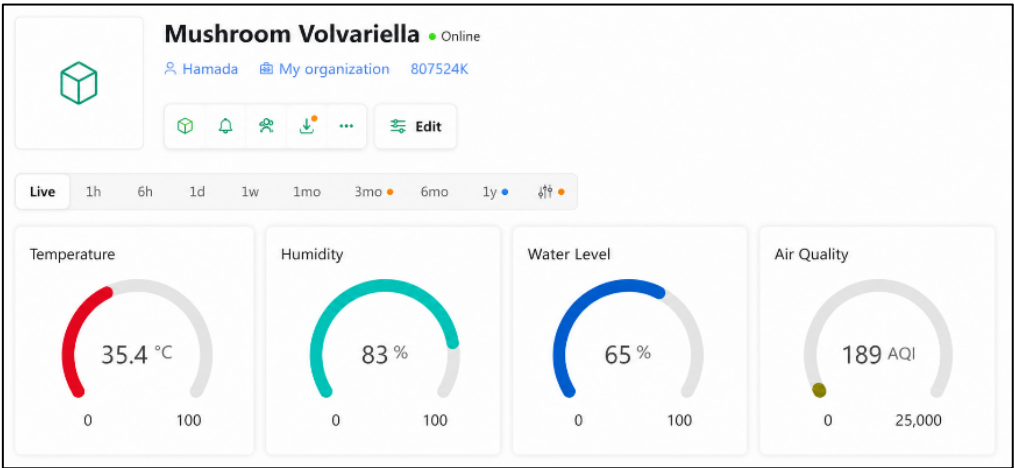


Figure 10. Web-based Blynk.Cloud monitoring dashboard for real-time environmental visualization.

30 The cloud platform not only showed environmental data but also provided an interactive user interface for the cultivation
31 chamber. Through the web and mobile applications, it was possible to observe the changes in the environmental conditions in
32 real time, to assess the cultivation status and to verify the functioning of the automated control subsystem. The mobile interface
33 also contributed to accessibility by providing information on cultivation directly on the mobile, and the graphical gauge widgets
34 made it easier to understand the data and allowed for the identification of deviations in the environment in a short time. The
35 visualization method is user-friendly, so the situation awareness is enhanced, and continuous monitoring can be done without
36 detailed analysis of raw sensor data. The remote monitoring feature also has a number of practical benefits in mushroom
37 farming. First, it decreases the labour required and makes it easier to save frequent physical inspections with continuous
38 oversight of the cultivation conditions. Second, it enables rapid identification of environmental abnormalities, allowing
39 corrective actions to be implemented before cultivation conditions deteriorate. Third, the provision of historic and current
40 environmental data provides better informed cultivation management and decision making. For commercial mushroom
41 growers, these advantages are especially important because of large growing areas, labour availability, or geographical distance,
42 among other factors, which may make it impractical to have constant supervision at the farm.
43 The proposed monitoring framework with the aid of clouds will offer a significantly more flexible and environmental
44 visibility compared to the conventional cultivation practices which are primarily based on manual observation. The achievement
45 of both web-based and mobile-based interfaces suggests the feasibility of incorporating the IoT communication technology into

intelligent mushroom cultivation systems. In summary, these results demonstrate real-time access to cultivation information with good reliability, in addition to complementing the automatic environmental regulation subsystem and enabling data-driven decisions. This capability of remotely monitoring cultivation conditions helps to optimize operation, create a stable environment, and promote the practice of smart mushroom farming.

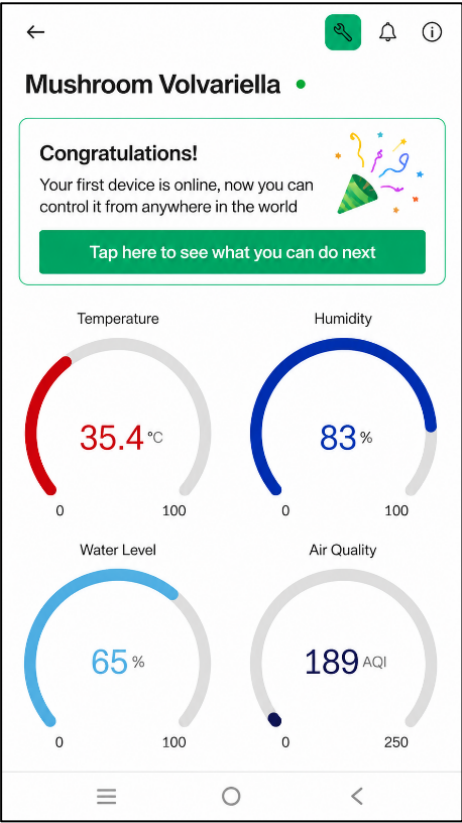


Figure 11. Mobile-based Blynk.Cloud monitoring interface for remote cultivation monitoring.

3.3 Mushroom Growth and Development Analysis

During the experimental period the growth progression of *Volvariella volvacea* was monitored to determine if the environmental conditions controlled by the automated control system were adequate for normal mushroom growth. Observations were made throughout the various stages of the cultivation, and it was observed that there was continuous change in the environment from mycelial colonization to the development of mature fruiting bodies which showed that the environmental regulation framework has maintained favourable growth conditions. At early colonization stage, dense white mycelial network developed throughout the substrate surface Fig. 12(a). The mycelium was uniformly distributed in the cultivation medium, indicating good adaptation of the fungus to the cultivation medium. Temperature, humidity and air-quality conditions were favourable for vegetative growth, with no visible contamination, and healthy cotton-like mycelial structures present. Environmental conditions were also conducive to reproductive structural development, as evidenced by the presence of early reproductive structures. When the pinhead structures appeared as in Fig. 12(b), it was a sign of the onset of the reproductive phase. This stage is considered to be one of the most environmentally sensitive stages in mushroom production and small changes in temperature, humidity, and/or gas concentration can inhibit the initiation of fruiting. The presence of more than one healthy pinhead, distributed throughout the cultivation substrate, indicated that the environmental control subsystem was able to provide sufficient microclimate stability during this critical period. The uniformity of pinhead formation is another indication that the automatic regulation system was effective in reducing environmental stress and in creating conditions that were favourable for the synchronous growth of mushrooms.

The next growth of the fruiting bodies and their maturation are shown in Fig. 13. Structural and size development of mushrooms was clearly observed compared to the previous developmental stage. Mature mushrooms had smooth surface morphology, uniform shape, and were healthy in colour, and with no sign of contamination, deformities, or physiological stress. These characteristics are typically linked to a good growing habitat and suggest good environmental conditions that have been provided during the entire cultivation period that have allowed for a healthy reproductive development. The proposed environmental management framework has been shown to be effective through the observed progression from mycelial colonization to mature fruiting-body formation. Based on these results, the incorporation of automated sensing and environmental regulation can be a part of the solution in ensuring cultivation consistency and minimize the uncertainty that is common in manual cultivation methods.

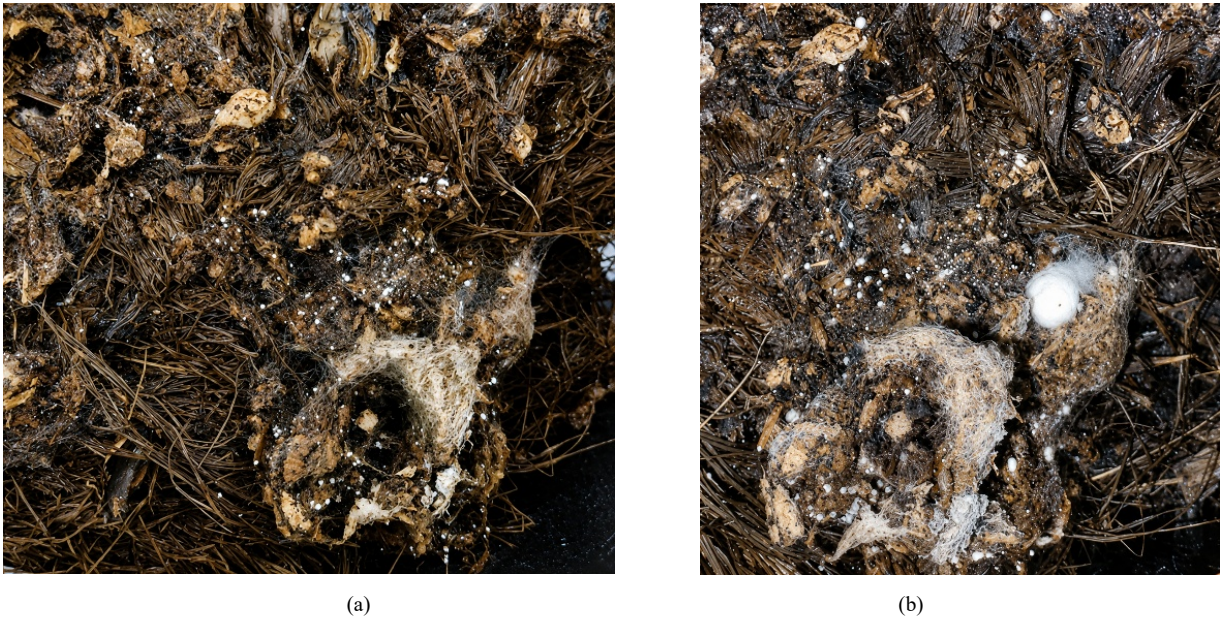


Figure 12(a) Initial mycelium colonization during the early cultivation stage. (b) Formation of early pinhead structures during the reproductive transition stage.



Figure 13. Development of mature fruiting structures during mushroom growth progression

9.3.4 YOLOv8-Based Mushroom Growth-Stage Detection Performance

The ability of the proposed YOLOv8 framework to automatically recognize the growth stage of *Volvariella volvacea* under actual cultivation conditions was assessed. Images were annotated into four classes: tiny, button, egg, and mature, and the model was trained for these classes. Figure 14 shows a sample of the process of annotation carried out with the Roboflow platform. In supervised deep-learning applications, the quality of the training labels directly affects the performance of the model and accurate annotation is crucial. The mushroom instances were classified into the different growth stages, and a bounding box was given to each instance of mushroom to allow the model to learn the spatial localization and different morphological features of mushroom in different growth stages. The distribution of the data set used for the model development is summarized in Figure 15. In total, 96 annotated images were captured and split into training set, validation set and testing set with 70, 20 and 6 images respectively. The number of images was relatively small, but mushroom size, developmental stage, orientation, clustering behaviour and background texture were varied to increase the diversity of images. These variations are crucial for the robustness of the model and for making it less sensitive to the environmental conditions under which it is deployed in the real world.

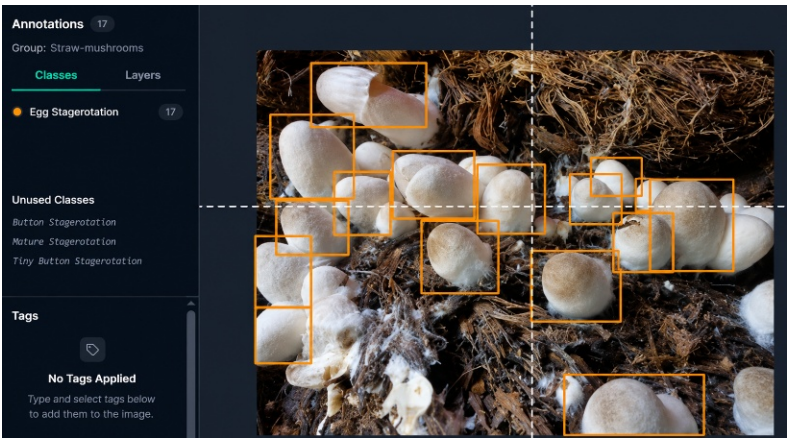


Figure 14. Sample mushroom image annotation using the Roboflow platform.

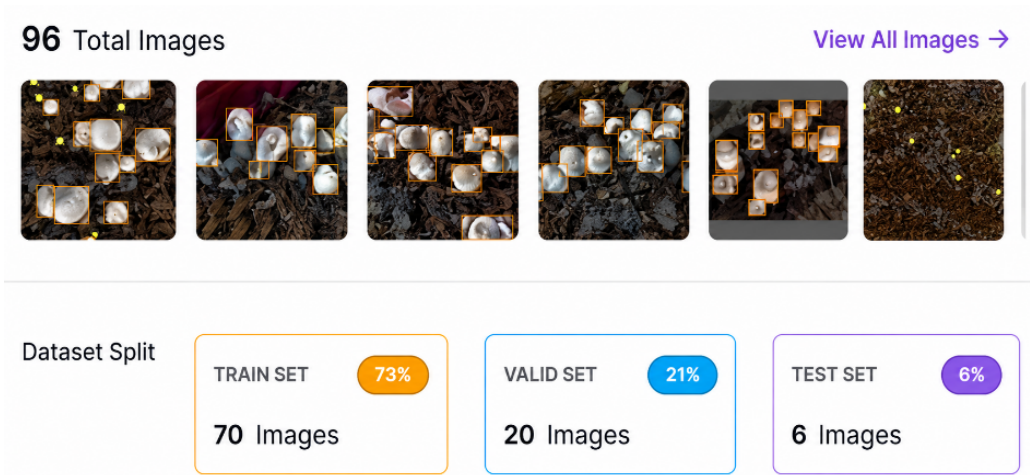


Figure 15. Distribution of training, validation, and testing datasets utilized for YOLOv8 model development.

The classification results of YOLOv8 model are shown in Fig. 16. Overall classification accuracy of the model was 93.12%, which indicates that the model has good performance in the growth stage recognition. The values of precision and recall were 100% for the mature stage, which shows that the model is perfectly precise and recall with full developed mushrooms. The tiny stage also showed good recognition performance with an F1-score of 97.5%, and the egg stage an F1-score of 89.3%. The majority of the misclassifications was between the button and egg stages, both of which are sequential developmental stages, and share similar morphology. This degree of confusion is recognized, as the size, shape and surface appearance of the tree can change over time making it more difficult to visually separate growth stages next to each other. However, the relatively small, misclassified samples suggest the model correctly learned discriminative features for each developmental stage. The effectiveness of the proposed classification framework is further supported by the confusion matrix as shown in Figure 16. The values on the diagonal are very high, suggesting that most samples were classified correctly into their growth stage categories. The small number of off-diagonal entries shows low misclassifications between different mushroom life stages, further highlighting the architecture's ability to classify similar mushroom life stages. The results also confirm the efficiency of the adopted dataset annotation technique and training configuration in this study. Figure 17 shows the typical training behaviour of the YOLOv8 model. The precision, recall and mean average precision (mAP) at an intersection-over-union (IoU) threshold of 0.5 was improved over 50 training epochs. The mAP@0.5 was 0.78, which was considered as good object-detection performance under realistic cultivation conditions. Meanwhile, the box loss and classification loss were steadily reduced throughout the training process, demonstrating steady learning behaviour and good parameter optimization. The smooth transition of the training curves suggests that the transfer-learning approach adopted, training regime, and hyperparameter choices allowed for effective model training without signs of significant overfitting or instability in the optimization process.

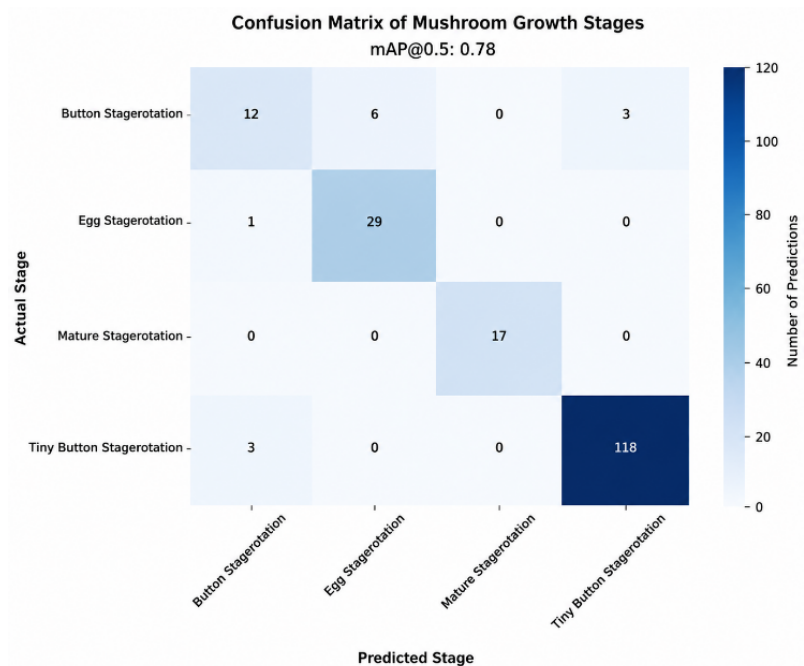


Figure 16. Confusion matrix of mushroom growth-stage classification performance.

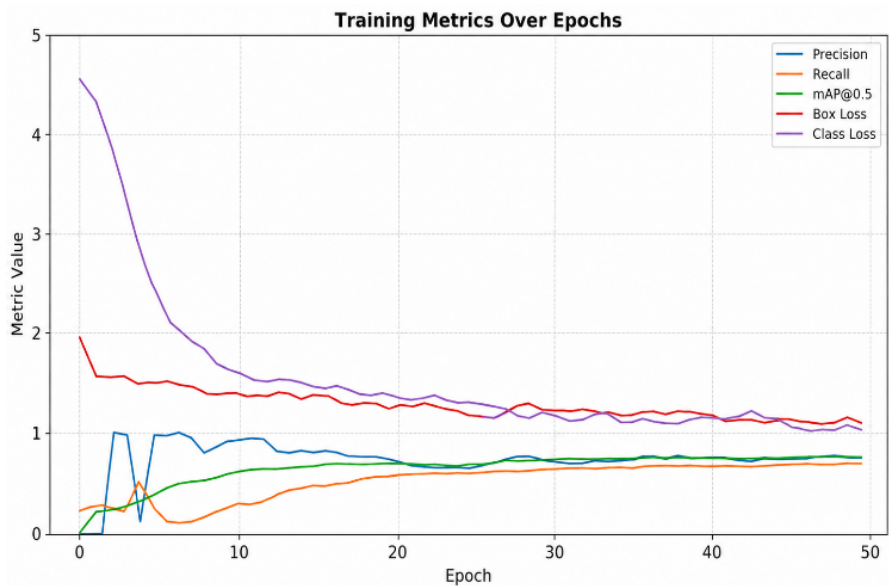


Figure 17. YOLOv8 training performance metrics over 50 training epochs.

The trained model also was evaluated in terms of its usefulness and tested by applying it in real time in the cultivation chamber, as shown in Fig. 18. From the live webcam images, multiple mushroom instances were successfully detected and classified both, with the confidence score of around 0.82 to 0.93. The detection performance was not influenced by illuminations, mushroom orientation, or overlap of mushrooms with the substrate texture. Correct localization and classification of mushrooms under these conditions, showcase the robustness of the proposed framework, while demonstrating the suitability for continuous cultivation monitoring. The use of visual analytics through the use of the YOLOv8 models, combined with the IoT for controlling the environment is a significant advancement from the mushroom farming method. The proposed system not only monitored mushroom development, but also constantly controlled the environmental conditions, unlike the traditional systems which heavily relied on manual inspection and experience-based assessment. This functionality will allow growers to get environmental and developmental information in real time, which will aid growers to make better cultivation and harvesting decisions. The achieved classification accuracy of 93.12% indicates that deep-learning-based computer vision can be used to support mushroom farming as an intelligent farm in a controlled environment.

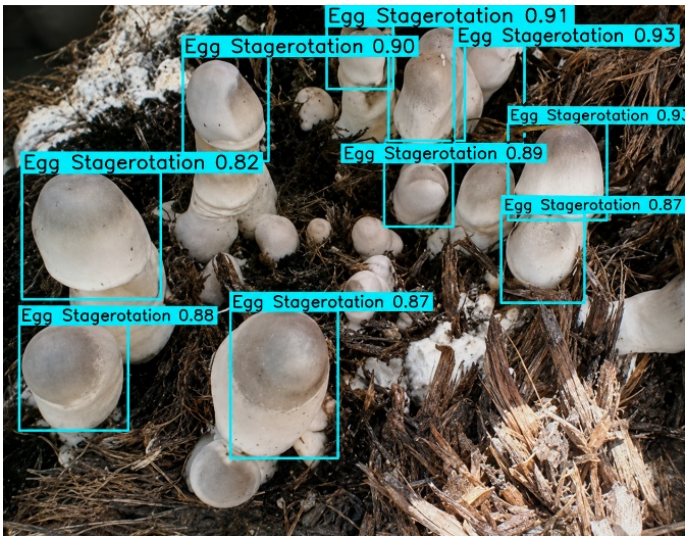


Figure 18. Real-time mushroom growth-stage detection using the trained YOLOv8 model.

While the findings were promising, there are a few caveats to consider. The first one was that the data were limited to 96 annotated images recorded from a single cultivation environment. Overall, the models performed well, but larger and more diverse datasets could enhance the generalization and robustness of the models. Second, the existing system is only about recognition of growth stages and not include disease detection, contamination monitoring or yield prediction. Lastly, the YOLOv8 model was run on a cloud computing platform instead of an edge computing platform. Future work could thus involve building up the dataset, adding more cultivation indicators, and implementing lightweight deep-learning models on edge devices with limited resources. Predictive analytics, disease surveillance, and yield estimation features could complement the proposed cultivation framework, providing additional intelligence and scalability. The overall experimental results revealed that the proposed intelligent growth management system can successfully integrate the environmental sensing, automatic environmental regulation, cloud monitoring, and deep learning recognition of the growth stage in the same cultivation platform. The environmental controlling system successfully met the environmental standards at about 85%, and the model YOLOv8 had an overall classification accuracy of 93.12% and mAP@0.5 value of 0.78 for four categories of mushroom growth stages. The results of this work confirm the feasibility for implementing the IoT-based automation and computer vision technologies to ensure cultivation consistency, decrease labor requirements and support precision mushroom cultivation in *Volvariella volvacea* cultivation.

IV. CONCLUSION

In this study, an intelligent growth management system for the growth of *Volvariella volvacea* mushrooms was designed and tested by introducing an integrated smart farming system for environment automation based on the IoT with growth stage recognition based on the You Only Look Once (YOLOv8) network. The proposed system overcame the drawbacks of traditional cultivation methods by effectively monitoring and controlling the cultivation environment as needed using sensors and actuators connected to the ESP32. By utilizing sensors and actuators connected to the ESP32, the proposed system was able to continuously monitor and automate the cultivation environment, such as temperature, relative humidity, carbon dioxide levels, and water-level conditions. Experimental results showed that the closed-loop cultivation conditions for environment control mechanism were maintained in the recommended range, and environmental compliance of ~85% was achieved during the whole cultivation period. At the same time, the YOLOv8 model is able to classify four stages of mushroom growth with an overall accuracy of 93.12% and a mean Average Precision (mAP@0.5) of 0.78, which is sufficient for the use of a reliable real-time system for mushroom growth monitoring in real cultivation conditions. Cloud monitoring through Blynk platform further enhanced access to the system and enabled remote supervision and reduced manual involvement. Overall, the findings demonstrate the potential of using IoT sensors for environmental management and deep learning visual analytics to improve mushroom production processes, such as environmental control, decision-making, and efficiency. The proposed framework is easily replicable and can be developed using affordable hardware and open-source technologies, allowing for scalable and practical solutions for precision agriculture applications and setting the groundwork for subsequent enhancements in autonomous farming systems, such as the integration of edge-AI, disease detection and forecasting, and commercialization on a large scale.

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